Attention-Guided Tabular-to-Image Representations for Deep Learning

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Abstract

Tabular data remains challenging for deep learning; tree-based ensembles often outperform neural architectures. One emerging direction transforms tabular data into image-like representations that enable convolutional neural networks (CNNs). However, most existing approaches rely on heuristic or unsupervised feature layouts not aligned with the predictive task.

We propose an attention-guided framework in which supervised feature attention is used to construct deterministic spatial representations for CNN learning. Using TabNet as an interpretable feature-attention backbone, we derive task-aware feature layouts from sequential sparse attention masks and project tabular samples into structured two-dimensional representations. The pipeline decouples representation learning from convolutional learning, enabling controlled analysis of how spatial organization influences downstream performance.

We implement and evaluate multiple tabular-to-image representations (naive reshaping, similarity-based layouts inspired by IGTD, and our attention-guided variants) alongside strong tabular baselines (gradient boosting ensembles, TabNet, and FT-Transformer-style architectures). Experiments on several benchmarks show that attention-guided spatialization provides stable, competitive performance while offering improved interpretability and task-aware spatial semantics. Although performance gains are generally modest, the results indicate that supervised feature attention provides a principled foundation for constructing meaningful spatial representations from tabular data.

1 Introduction

Deep learning has achieved remarkable success in domains with strong spatial or sequential structure, such as computer vision and natural language processing. In contrast, tabular data remains a challenging setting for neural architectures, with tree-based ensemble methods such as XGBoost, LightGBM, and CatBoost often outperforming deep learning models in both accuracy and robustness [Grinsztajn et al., 2022]. This gap has motivated two complementary research directions: the development of specialized neural architectures for tabular learning and the design of alternative representations that make tabular data more suitable for deep models.

One increasingly explored direction involves transforming tabular data into image-like representations so that CNNs can be applied [Zhu et al., 2021, Sharma et al., 2019, Lin and Wu, 2026, Gómez-Martínez et al., 2026, Alenizy and Berri, 2025, Liang et al., 2025, Serra Neto et al., 2022]. Existing tabular-to-image methods typically organize features according to statistical similarity, dimensionality reduction, or predefined spatial heuristics [Sharma et al., 2019, Bazgir et al., 2019, Zhu et al., 2021, Alenizy and Berri, 2025]. While these approaches allow CNN-based learning on structured representations, the resulting layouts often lack direct alignment with the supervised prediction task and may provide limited semantic interpretability.

We argue that spatial organization should be guided by task-aware feature relevance rather than by unsupervised similarity structure or arbitrary ordering. Instead of externally imposing geometry, we derive spatial layouts from an interpretable tabular model trained on the target task. In particular, we leverage TabNet's sequential sparse attention mechanism [Arik and Pfister, 2021], which produces structured, task-dependent feature-selection masks across multiple decision steps. These attention masks capture supervised feature relevance and higher-order grouping behavior, which can be aggregated into deterministic spatial priors for image construction.

Based on these attention patterns, we generate two-dimensional feature layouts that preserve supervised feature structure while remaining compatible with standard CNN architectures. Importantly, the proposed pipeline decouples spatial representation learning from convolutional learning: the tabular attention model is trained independently, its learned structure is frozen, and CNNs are trained exclusively on the generated image representations. This separation enables controlled analysis of how layout geometry and feature organization affect downstream performance.

Beyond the proposed attention-guided layouts, this work also implements and evaluates several alternative tabular-to-image strategies under a unified experimental protocol, and compares them against strong tabular baselines including XGBoost, LightGBM, CatBoost, TabNet, and FT-Transformer-style architectures. The framework is related to spatialization pipelines such as MOL [Serra Neto et al., 2022], but derives spatial organization directly from supervised attention rather than from metaheuristic search.

To support transparency, we provide an interactive application that allows inspection of feature attention, spatial layouts, and generated image representations throughout the pipeline.

Figure 1 summarizes the proposed framework. TabNet is first trained on the original tabular task to learn supervised feature attention and grouping patterns. The learned attention structure is then frozen and used to construct

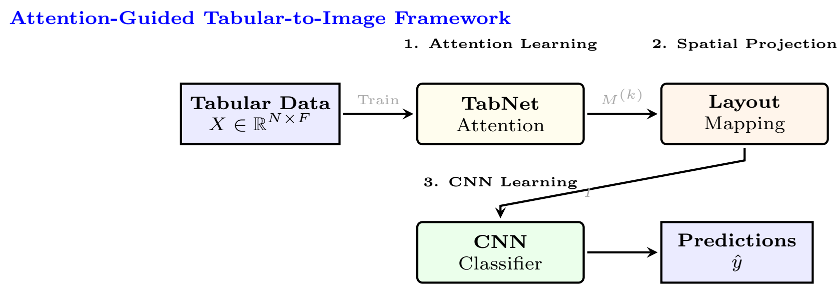

Figure 1: Overview of the proposed attention-guided tabular-to-image framework. TabNet is used to learn supervised feature attention, which is transformed into fixed spatial layouts for CNN-based learning. The pipeline decouples representation learning from spatial projection.

deterministic two-dimensional image representations for CNN training. No gradients flow from the CNN back to the tabular model, keeping representation learning and spatial projection fully decoupled.

2 Contributions

This work makes the following contributions:

- We propose an attention-guided tabular-to-image framework that derives deterministic spatial layouts from TabNet's supervised sparse attention mechanism, decoupling representation learning from CNN training.
- We introduce task-aware feature-to-grid mapping strategies (StepRow, StepSparse, PackedRow, PackedCol and AttentionMap) that transform tabular samples into structured CNN-compatible image representations while preserving interpretable feature organization.
- We provide an open-source pipeline with an interactive visualization tool for inspecting attention, spatial layouts, and generated images, facilitating reproducibility and interpretability studies.

3 Related Work

3.1 Deep Learning for Tabular Data

Despite extensive research, deep neural networks have not consistently surpassed tree-based methods on tabular benchmarks [Grinsztajn et al., 2022, Gorishniy et al., 2021, Borisov et al., 2024]. Tree-based ensemble methods such as XGBoost, LightGBM, and CatBoost remain dominant due to their strong inductive biases and robustness. Several architectures introduce inductive biases tailored to tabular structure, including NODE [Popov et al., 2019], TabTransformer [Huang et al., 2020], FT-Transformer [Gorishniy et al., 2021], and TabNet [Arik and Pfister, 2021]. Among these, TabNet is notable for its sparse, step-wise attention mechanism that provides built-in interpretability by explicitly selecting features at each decision step.

3.2 Tabular-to-Image Representations

A parallel line of work converts tabular data into images to exploit CNNs. IGTD [Zhu et al., 2021] arranges features to preserve pairwise similarity, while DeepInsight [Sharma et al., 2019] maps features to fixed spatial grids based on statistical structure. These methods improve over naive reshaping, but the resulting layouts are typically heuristic or similarity-based, not explicitly aligned with the predictive task [Alenizy and Berri, 2025]. Structured transformation pipelines such as MOL [Serra Neto et al., 2022] combine feature selection, metaheuristic optimization, and convolutional learning to construct spatialized tabular representations. More recent work emphasizes preserving feature order or spatial coherence [Alenizy and Berri, 2025]. Despite growing interest, the lack of standardized large-scale benchmarks and unified evaluation protocols makes direct comparison difficult [Liu et al., 2026, Gómez-Martínez et al., 2024].

3.3 Hybrid and Knowledge-Guided Learning

Hybrid approaches combine tabular models and deep networks via teacher-student learning or representation distillation [Henrich et al., 2026, Dissanayake and Dutta, 2025]. Our work differs by using an interpretable tabular model solely to define the spatial structure of the image representation, after which the model is frozen and discarded. The geometry induced by feature attention can then be analyzed independently of tabular prediction performance.

4 Method

4.1 Overview

The pipeline consists of three stages: (i) supervised attention learning on tabular data, (ii) spatial projection of features into image-like representations, and (iii) CNN-based learning on the generated images. First, a TabNet model is trained on the original tabular dataset to learn task-dependent feature relevance through sparse sequential attention masks. The learned attention structure is then frozen and used to construct deterministic spatial layouts. Finally, convolutional neural networks are trained on the resulting image representations.

All tabular-to-image methods are evaluated using the same preprocessing pipeline, train-test splits, and CNN backbone architecture in order to isolate the effect of spatial layout construction. No gradients flow from the CNN stage back to the attention model, keeping representation learning and convolutional learning fully decoupled.

4.2 Preprocessing

Numerical features are standardized using training-set statistics, categorical variables are encoded via label or one-hot encoding, and missing values are imputed. All operations are fitted on training data only and applied to validation/test partitions. The same pipeline is shared across all methods.

4.3 Attention-Guided Spatial Representation

Let $X \in \mathbb{R}^{N \times F}$ be a tabular dataset with N samples and F features. A TabNet model with K decision steps is trained on the original classification task. For every feature f_j, TabNet produces a soft step-assignment vector

$$a_j = (a_{j1}, \ldots, a_{jK}), \qquad \sum_{k=1}^{K} a_{jk} = 1,$$

where a_{jk} is the probability that feature j is attended to at decision step k. A global importance score s_j is computed as the sum (or average) of the step-wise attention weights,

$$s_j = \sum_{k=1}^{K} a_{jk}.$$

For spatial layouts that require a hard step assignment, each feature is mapped to the step where it receives maximum attention,

$$k_j^* = \arg\max_k a_{jk}.$$

Features are grouped by their dominant step k_j^* and, within each group, sorted in descending order of global importance s_j.

The two quantities a_j and s_j are extracted once from the trained TabNet model and then frozen. All subsequent image generation relies on this fixed attention structure, completely decoupling the representation step from the CNN training stage.

4.4 Tabular-to-Image Mapping Strategies

The framework supports multiple feature-to-image mapping strategies to study how spatial organization affects CNN learning.

Naive Reshaping Features are placed sequentially into a two-dimensional grid using row-major ordering, with zero-padding for empty positions. This preserves feature order but encodes no semantic relationships.

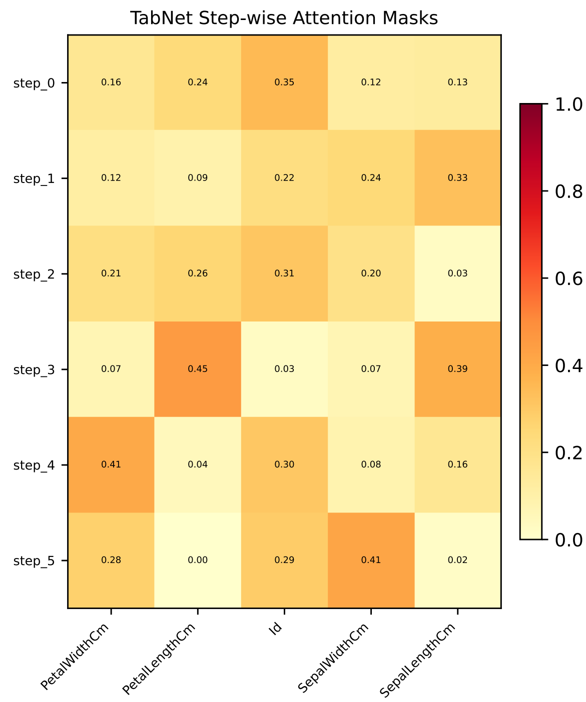

Figure 2: Illustration of the masks for the Cancer dataset, showing how different steps specialise to different feature groups.

IGTD-Inspired Layouts We implement a lightweight similarity-based layout: pairwise feature similarities are computed using correlation-based distances, and multidimensional scaling (MDS) projects features into a 2D latent space; features are then assigned to grid locations according to spatial proximity, approximately preserving similarity relationships, but not yeilding identic results to the original.

Attention-Guided Layouts The attention-guided layouts exploit the feature attribution information produced by TabNet to construct deterministic image representations. Let F denote the number of features and K the number of TabNet decision steps.

For layouts based on dominant-step assignments, each feature f_j is assigned to the decision step

$$k_j^* = \arg\max_k a_{jk},$$

where a_{jk} denotes the normalized attention weight of feature j at step k. Within each step, features are ordered according to their global importance score s_j. The resulting ordered feature groups are then mapped onto a two-dimensional image grid through a deterministic coordinate mapping

$$\phi : f_j \to (r_j, c_j).$$

Different layouts impose different spatial organizations:

- **AG-T2I-StepRow** preserves the TabNet decision structure by allocating one image row to each decision step. Features assigned to the same dominant step are placed consecutively from left to right in descending importance order. The resulting image height equals the number of active decision steps, while the width corresponds to the largest step group.
- **AG-T2I-StepSparse** also allocates one row per decision step but reserves a fixed number of columns for every step. Empty positions are preserved whenever fewer features are assigned to a step, explicitly maintaining step boundaries and improving interpretability of the spatial representation.
- **AG-T2I-PackedRow** discards the step grouping after feature ranking and densely packs all retained features into a compact row-major grid ordered by decreasing global importance. This strategy maximizes spatial compactness and local feature density for CNN processing.
- **AG-T2I-PackedCol** is the transposed version of AG-T2I-PackedRow. Features are packed according to the same global importance ordering, but the grid orientation is exchanged, producing a column-major arrangement.
- **AG-T2I-AttentionMap** constructs an image of size $K \times F$ where each pixel is computed as

  $$I(k, j) = \tilde{a}_{jk} \cdot \tilde{s}_j \cdot x_j.$$

  Here $\tilde{a}_{jk}$ is the normalised attention weight of feature j at decision step k, $\tilde{s}_j = s_j / \max_l s_l$ is the normalised global importance of feature j, and x_j is the feature value of the given sample. In contrast to the coordinate-based layouts, this representation directly encodes the interaction between the learned attention structure and the actual sample values.

4.5 CNN-Based Learning and Configuration

All tabular-to-image representations are evaluated using the same lightweight CNN architecture: three convolutional blocks with ReLU activations and max-pooling, followed by fully connected layers. The network is trained with Adam optimizer ($lr = 10^{-3}$, batch size 32) and early stopping. Dropout is applied, especially on smaller datasets such as Glass. Since spatial layouts are fixed before CNN training, the convolutional model learns exclusively from the induced spatial structure.

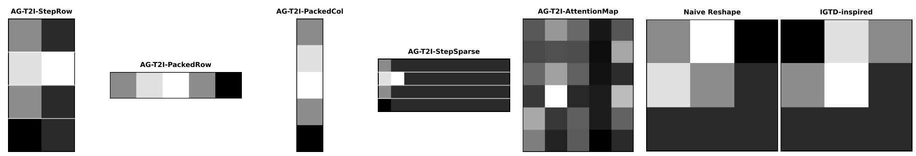

Figure 3: Feature layout comparison for the Cancer dataset. From left to right: AG-T2I-StepRow, AG-T2I-PackedRow, AG-T2I-PackedCol, AG-T2I-StepSparse, AG-T2I-AttentionMap, Naive Reshape and IGTD-inspired layout. Features are color-coded by original variable groups; the attention-guided layout places semantically related features closer together, providing a more structured spatial organization for CNN processing.

4.6 Tabular Learning Baselines

Strong tabular baselines include gradient boosting ensembles (XGBoost, LightGBM, CatBoost), TabNet, and an FT-Transformer-style architecture implemented with lightweight transformer encoder blocks.

5 Experimental Protocol

5.1 Datasets and Compared Methods

We evaluate on multiple public tabular classification benchmarks (Table 1) covering binary and multiclass tasks with heterogeneous feature types. Compared methods include classical ensemble models, neural tabular architectures, and tabular-to-image representations (naive reshaping, IGTD-inspired layout, and the three AG-T2I variants). All methods share identical preprocessing and train-test splits.

5.2 Controlled Evaluation Procedure

For attention-guided layouts, TabNet is first trained on the original tabular task; the learned attention structure is frozen and reused during image generation. CNN training is performed exclusively on the generated image representations, with no gradient propagation back to the tabular model. All experiments use fixed preprocessing, shared data splits, and identical CNN settings.

5.3 Evaluation Metrics and Statistical Note

Performance is evaluated using classification accuracy and macro-F1 score, averaged over multiple runs with standard deviations reported. Given the modest number of datasets, formal statistical testing has limited power; we therefore interpret differences qualitatively, reporting means and standard deviations to convey variability.

Table 1: Summary of benchmark datasets.

Dataset	Samples	Features	Classes
Cancer (Can)	699	9	2
Diabetes (D)	768	8	2
Glass (Gl)	214	9	6
Thyroid (T)	7,200	21	3
Card (Car)	690	51	2
Gene (Ge)	3,175	120	3
Heart (He)	920	35	2
Horse (Ho)	364	58	3
Soybean (S)	683	82	2
KDD	494,200	42	2
Poker Hand (PH)	1,025,010	18	2
Cardiac Pathology (CP)	9,484	54	7
Forest Cover Type (FCT)	581,012	11	10

6 Results

This section evaluates whether attention-guided spatial representations provide an effective and interpretable foundation for CNN-based learning on tabular data, focusing on comparison against existing tabular-to-image representations, strong tabular baselines, and the influence of supervised spatial organization.

6.1 Predictive Performance

Tables 2 and 3 reports mean classification accuracy across repeated stratified 5-fold cross-validation runs. AG-T2I variants are defined as in Section 4.4.

Table 2: Classification accuracy (%) – Part 1: Cancer through Soybean. Best in bold, second underlined, third in gray.

Model	Can	D	Gl	T	Car	Ge	He	Ho	S
XGBoost	95.71	74.73	75.23	99.46	...	...	...	...	...
LightGBM	96.42	74.34	**79.90**	**99.67**	...	...	...	...	...
CatBoost	96.85	<u>76.69</u>	<u>78.96</u>	<u>99.54</u>	...	...	...	...	...
TabNet	**97.42**	**77.99**	65.87	93.63	...	...	...	...	...
FT-Transformer	96.14	73.56	72.39	99.06	...	...	...	...	...
IGTD-inspired	96.57	74.87	56.51	98.18	...	...	...	...	...
Naive Reshape	<u>97.28</u>	73.95	47.18	98.03	...	...	...	...	...
AG-T2I-StepRow	96.94	72.94	50.28	96.77	...	...	...	...	...
AG-T2I-StepSparse	96.88	75.67	61.59	97.21	...	...	...	...	...
AG-T2I-PackedRow	96.71	72.28	40.46	95.66	...	...	...	...	...
AG-T2I-PackedCol	96.88	75.67	61.59	97.21	...	...	...	...	...
AG-T2I-AttentionMap	96.88	75.67	61.59	97.21	...	...	...	...	...

Table 3: Classification accuracy (%) – Part 2: KDD through Forest Cover Type. Best in bold, second underlined, third in gray.

Model	KDD	PH	CP	FCT
XGBoost	...	...	...	...
LightGBM	...	...	...	...
CatBoost	...	...	...	...
TabNet	...	...	...	...
FT-Transformer	...	...	...	...
IGTD-inspired	...	...	...	...
Naive Reshape	...	...	...	...
AG-T2I-StepRow	...	...	...	...
AG-T2I-StepSparse	...	...	...	...
AG-T2I-PackedRow	...	...	...	...
AG-T2I-PackedCol	...	...	...	...
AG-T2I-AttentionMap	...	...	...	...

Overall, attention-guided tabular-to-image representations are competitive with existing spatialization approaches and, in several cases, improve upon naive reshaping and similarity-based layouts. However, tree-based ensemble methods remain highly effective. Across datasets, performance differences are often moderate, consistent with the limited performance headroom noted in prior work [Grinsztajn et al., 2022].

6.2 Dataset-Level Insights

Cancer Most methods perform strongly; TabNet achieves the highest accuracy, with naive reshaping and AG-T2I variants very close, indicating that simple spatial structure already suffices for this dataset.

Diabetes AG-T2I-StepSparse achieves the best result among image-based methods and approaches the top tree ensembles, suggesting that supervised feature grouping becomes more valuable as task difficulty increases.

Glass All image-based methods suffer from the extremely small sample size, but AG-T2I-StepSparse substantially improves over naive reshaping and IGTD, confirming that attention-guided organization remains beneficial even in low-data regimes. Densely packed layouts degrade strongly.

Thyroid Tree-based models dominate; among image-based approaches, AG-T2I-StepSparse again provides the best performance, demonstrating that explicit step-wise separation aids CNN learning more than dense packing.

6.3 Effect of Layout Strategy

StepSparse consistently yields the most stable results across datasets, preserving explicit separation between TabNet decision steps and maintaining localized feature groupings that facilitate convolutional learning. StepRow, while competitive, can create elongated spatial regions less suited to local filters. Packed layouts perform worst, indicating that discarding step-wise structure hurts CNN performance more than any gains from spatial compactness. These trends underscore that the primary benefit comes from supervised feature organization, not arbitrary geometric arrangement.

6.4 Interpretability

A key advantage of the proposed framework is that each spatial location is directly associated with a specific tabular feature, its supervised importance score, and a dominant TabNet decision step. Inspection of generated layouts and CNN saliency maps suggests that highly attended feature groups form coherent spatial regions that are subsequently exploited during convolutional learning, providing explicit semantic traceability absent from heuristic or similarity-based layouts.

7 Discussion

This study demonstrates that supervised feature attention provides a meaningful foundation for constructing tabular-to-image representations. Attention-guided layouts consistently improve over naive reshaping and generally outperform similarity-based spatialization, especially on more challenging datasets. The comparison among AG-T2I variants reveals that preserving step-wise feature organization is more important than maximizing spatial compactness, as step-sparse layouts regularly surpass densely packed alternatives.

At the same time, the results confirm that strong tree-based ensemble methods remain difficult to surpass on structured tabular benchmarks. The primary contribution of this work is therefore not to claim state-of-the-art predictive performance, but to offer a principled, interpretable mechanism for spatializing tabular data and to provide a controlled framework for studying how spatial organization influences convolutional learning.

Several limitations should be noted. The experiments are limited to a small set of benchmark datasets; large-scale, high-dimensional industrial datasets may exhibit different behavior. The attention-guided spatial structure is derived from a single tabular model and may be suboptimal for CNN learning; jointly optimizing layout and representation could yield further improvements. Moreover, the spatialization process introduces an additional preprocessing stage, increasing pipeline complexity relative to direct tabular learners. Finally, while attention-guided layouts improve interpretability, CNNs still generally underperform tree-based models, and the observed performance gains are modest. Nevertheless, the controlled separation of attention learning and CNN training allows clear figures/generated_image_examples.pdf

Figure 4: Generated image examples for one sample from the Cancer dataset under three layout strategies. Left: Naive Reshape; Center: IGTD-inspired; Right: AG-T2I-StepSparse. The attention-guided layout groups related features into contiguous spatial regions, creating a more structured input for the CNN.

investigation of how layout geometry affects representation quality—an aspect often obscured in end-to-end approaches.

The interactive application that accompanies this work further highlights the practical interpretability of the pipeline, enabling direct inspection of attention, spatial layouts, and generated images.

8 Conclusion

We introduced an attention-guided tabular-to-image framework that leverages supervised feature attention from TabNet to construct deterministic spatial representations for CNN-based learning. The proposed method decouples representation learning from convolutional learning, enabling a controlled study of spatial organization. Experiments across multiple benchmarks demonstrate that attention-guided layouts provide competitive, interpretable CNN-compatible representations, with step-sparse variants yielding the most stable behavior.

Although not surpassing strong tree-based ensembles, the framework offers a transparent mechanism for spatializing tabular data and for investigating how supervised feature structure influences convolutional learning. Future work may explore differentiable spatialization, adaptive graph-based layouts, vision transformer architectures, and hybrid CNN-transformer models operating directly on attention-guided representations.

A Implementation Details

A.1 Interactive Pipeline Tool

A Streamlit application allows users to execute and inspect the full pipeline without writing code. The tool supports dataset loading, preprocessing configuration, TabNet training, spatial layout selection, CNN training, and evaluation, all orchestrated via subprocess calls to the same command-line scripts. The dashboard visualises feature importance, attention masks, feature-to-pixel assignments, and generated image grids.

A.2 Key Software Components

The framework comprises a collection of Python scripts:

- `run_preprocessing.py` – loads raw data, applies configurable cleaning, scaling, encoding, and saves train/test splits.
- `train_tabnet.py` – trains TabNet and exports attention masks, feature importance, and step assignments.
- `unified_layouts.py` – defines deterministic layout strategies (step-row, packed, step-sparse).
- `tabnet_image_builder.py` – projects tabular features into 2D images using a chosen layout.
- `cnn_model.py` – provides a lightweight CNN (`TabNetCNN`) that adapts to image dimensions.
- `train_cnn.py` / `evaluate_cnn.py` – train the CNN on generated images and compute evaluation metrics.
- `api.py` – command-line entry point for single runs, random search, and Bayesian optimisation.

All modules are available at `https://github.com/...` (to be inserted).

A.3 Attention-to-Grid Algorithm

Algorithm 1 formalises the deterministic mapping from TabNet attention to a 2D grid.

Algorithm 1 Attention-guided tabular-to-image mapping

Require: TabNet masks $M^{(k)}$, feature dimension F, grid size $H \times W$
Ensure: Image representation $I \in \mathbb{R}^{H \times W}$
1: Compute importance scores $s \leftarrow \frac{1}{K} \sum_{k=1}^{K} M^{(k)}$
2: Assign each feature to step $k^* \leftarrow \arg\max_k M^{(k)}$
3: Sort features by (step, importance) in descending order
4: Fill grid sequentially (row-major) with ordered features; pad empty cells with 0
5: **return** I

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